Supplementary Material for "TRACE-R: Telemetry-Routed Adaptive Corruption-Expert Restoration for Pavement Inspection"

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TABLE I
VALIDATION-FIXED TRACE-R ROUTING CONSTANTS.

Quantity	Value
Motion threshold t_M	0.180
Defocus threshold t_D	0.200
Low-light threshold t_L	0.385
Reliability threshold q_0	0.500
DFPIR weight, low-light route	0.400
DFPIR weight, mixed route	0.075
Gyroscope full scale	4.0

I. Capture Interface

The public packet contains 82 normalized values. Entries 1–33 are eleven three-axis gyroscope samples and entries 34–66 are eleven three-axis accelerometer samples, both centred on exposure. The remaining context entries are, in order: log exposure, log ISO, analog gain, focal length, aperture, focus-error proxy, autofocus confidence, rolling-shutter readout, vehicle speed, vehicle yaw rate, timestamp offset, IMU noise, camera reliability, IMU reliability, vehicle reliability, and global availability. Camera, IMU, and vehicle masks are independent.

The deterministic map is implemented by `observable_code_from_packet`. The reported router is implemented by `TRACERExpertFusion.route`. Neither function accepts a dataset identifier, scenario label, target image, detector label, or hidden synthetic kernel. Controlled sidecars use the same packet order as measured KITTI and CRID records.

Motion uses DFPIR and defocus uses InstructIR. The low-light blend is 0.40 DFPIR plus 0.60 DeMoE; the mixed blend is 0.075 DFPIR plus 0.925 DeMoE. The all-unavailable route uses DeMoE. These coefficients are stored by `TRACERPolicy`; they are not recomputed for an evaluation image.

II. Matched Adaptation and Checkpoints

Every controlled restorer receives 70 total target-domain epochs, 4,096 effective optimizer updates, an effective batch size of four, and source-balanced IVCNZ/PCM sampling. AdamW uses an initial learning rate of 10^{-5} and weight decay 10^{-4}. The common objective has Charbonnier coefficient 1.0, gradient coefficient 0.15, frozen-detector feature coefficient 0.20, and frozen supervised detector coefficient 0.10. Detector features are taken from layers `model.2` and `model.4` with relative weights 0.5 and 1.0. Detector terms use a one-epoch linear warm-up. Validation mAP50 selects checkpoints; the training script does not evaluate a test split.

TRACE-R does not receive an additional optimization budget. It composes the selected DFPIR, InstructIR, and DeMoE checkpoints in Table III. NAFNet and FiLM-NAFNet are trained under the same common objective but are not included in the expert bank because their validation accuracy is lower.

III. Controlled Results

Table IV gives all eight controlled condition results. TRACE-R wins or ties seven conditions. DeMoE is higher on PCM mixed degradation by 0.0006 before rounding. This exception is retained in the ledger and is not rounded into a strict TRACE-R win.

Figure 1 is the complete IVCNZ comparison used in the main paper. TRACE-R and DeMoE tie on recovered ground-truth count for this frame. The example is retained because it was selected by the fixed recovery rule and illustrates the aggregate low-light behavior without implying a strict image-level win.

IV. Measured KITTI Transfer

The KITTI study uses drive-disjoint images and measured OXTS motion. Its checkpoint belongs to the earlier RMR-P arXiv model; it is not presented as a TRACE-R expert-router result. The purpose is to verify that the 82-value interface can be populated from a public measured motion record.

V. CRID Field Protocol

CRID images remain at 4752×3168. No synthetic degradation is added. The restoration expert is adapted on unlabelled CRID imagery and paired synthetic corruptions without reading the 46 manual field labels. The detector and policy operate at native image coordinates.

The earlier 12-frame temporal segment fixes the complete field policy in Table VIII. The evidence statistic is the sum of squared detector confidences above 0.05. A zero relative margin admits restored evidence only when its score is strictly higher than the native score. Restored predictions below 0.10 are omitted before same-family fusion, and duplicates are removed at IoU 0.15. The detector operating threshold is 0.20. These are the exact values in `frozen_policy_before_supportive_test.json`.

The subsequent 13-frame temporal block contains 49 ground-truth objects. Table IX repeats the complete ranked comparison. The block was inspected during prior development and is therefore supportive rather than newly sealed evidence.

TABLE II
MATCHED TARGET-DOMAIN ADAPTATION AND SELECTION BUDGET.

Method	Epochs	Optimizer steps	Selection endpoint
NAFNet	70	4,096	joint validation mAP50
FiLM-NAFNet	70	4,096	joint validation mAP50
InstructIR	70	4,096	joint validation mAP50
DFPIR	70	4,096	joint validation mAP50
DeMoE	70	4,096	joint validation mAP50
TRACE-R router	matched experts	0	joint validation mAP50

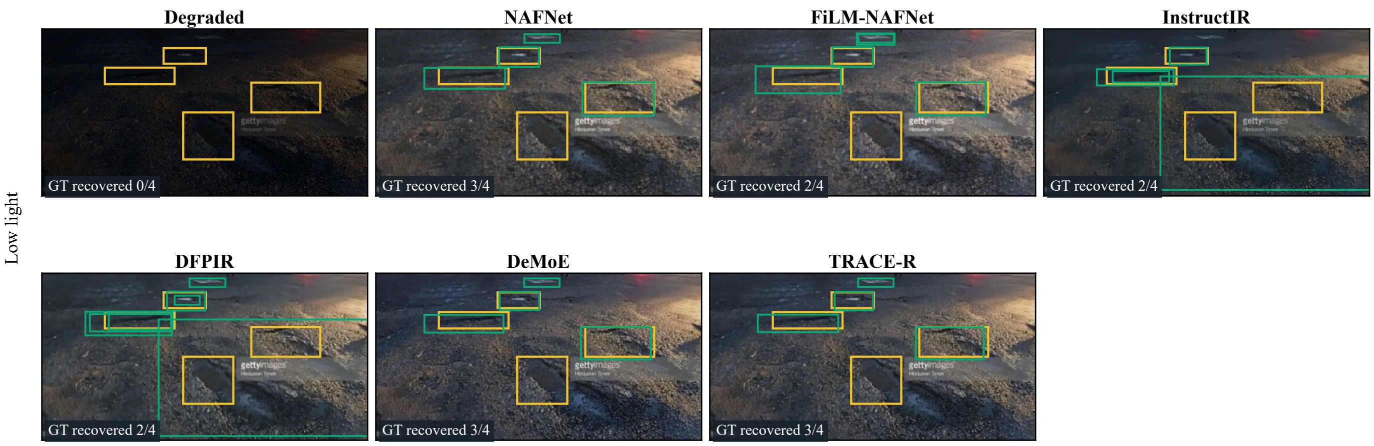

Fig. 1. IVCNZ validation example with all controlled comparators. Yellow: annotation; green: detector prediction.

Fig. 2. Drive-disjoint KITTI paired transfer. The panel preserves the executed RMR-P checkpoint name.

TABLE III
TRACE-R EXPERT CHECKPOINTS.

Expert	Adaptation epochs	SHA-256 prefix
DeMoE	70	`743f106d380bd8db`
DFPIR	70	`cca8f4b00a2e6468`
InstructIR	70	`4d97860f360bf21c`

VI. Evidence and Reproducibility Boundaries

The IVCNZ and PCM results are sequence/source-disjoint validation evidence. Earlier development opened the original test partitions, so those partitions are excluded from the reported claim. KITTI uses an earlier arXiv checkpoint. CRID uses a temporally earlier calibration segment and a non-overlapping but previously inspected evaluation segment. These boundaries are encoded in the provenance ledger and are not inferred from directory names.

The public release includes packet builders, matched training commands, expert-routing code, detector adapters, route constants, checkpoint hashes, qualitative-selection manifests, and the frozen result ledger. Historical file aliases remain only to reproduce old manifests. Paper-facing code and comments

TABLE IV
CONDITION-LEVEL MAP50 ON IVCNZ AND PCM.

Method	IVCNZ				PCM			
	Mot.	Def.	Low	Mix.	Mot.	Def.	Low	Mix.
NAFNet	0.374	0.297	0.542	0.385	0.169	0.142	0.212	0.039
FiLM-NAFNet	0.375	0.296	0.544	0.387	0.170	0.146	0.221	0.047
InstructIR	0.510	**0.523**	0.546	0.433	0.316	0.297	0.267	0.081
DFPIR	**0.543**	0.498	**0.555**	0.415	**0.345**	**0.302**	0.296	0.138
DeMoE	0.536	0.509	0.543	0.480	0.326	0.289	0.308	**0.212**
TRACE-R	**0.543**	**0.523**	0.553	**0.482**	**0.345**	0.297	**0.311**	0.211

TABLE V
PAIRED FIDELITY OVER ALL FOUR CONTROLLED CONDITIONS.

Method	IVCNZ		PCM	
	PSNR	SSIM	PSNR	SSIM
Degraded input	16.70	0.472	18.23	0.613
InstructIR	21.20	0.656	24.52	0.807
DFPIR	24.06	0.707	26.32	0.826
DeMoE	23.78	0.715	26.13	0.838
TRACE-R	24.12	0.726	26.46	0.842

TABLE VI
TRACE-R PACKET INTERVENTIONS.

Packet supplied at inference	IVCNZ	PCM	Joint
Aligned packet	0.525	0.291	0.408
All sensor groups unavailable	0.466	0.236	0.351
Wrong-condition packet	0.451	0.213	0.332

TABLE VII
DRIVE-DISJOINT KITTI TRANSFER WITH MEASURED OXTS TELEMETRY.

Method	Capture input	PSNR	SSIM	ΔPSNR
Degraded input	none	26.06	0.845	0.00
NAFNet	none	26.03	0.848	-0.03
FiLM-NAFNet	OXTS	25.96	0.846	-0.10
RMR-P	unavailable packet	26.82	0.865	+0.76
RMR-P	OXTS	**26.86**	**0.865**	**+0.79**

TABLE VIII
FROZEN CRID DETECTOR-EVIDENCE POLICY.

Quantity	Value
Evidence statistic	$\sum_{p_i \geq 0.05} p_i^2$
Relative gate margin τ	0.000
Restored-prediction floor	0.100
Same-family fusion IoU	0.150
Detector operating threshold	0.200

use TRACE-R, and all numerical tables in the manuscript are generated from the frozen ledger or the frozen CRID summaries.

TABLE IX
CRID NATIVE-IMAGE DETECTION ON THE 13-FRAME TEMPORAL EVALUATION BLOCK.

Input	AP@.10	AP@.50	Mean
Raw image	0.516	0.446	0.481
NAFNet	0.559	0.474	0.516
DFPIR	0.550	0.452	0.501
DeMoE	0.551	0.452	0.502
InstructIR	0.542	0.442	0.492
TRACE-R	**0.560**	**0.484**	**0.522**

Fig. 3. All 13 CRID temporal-evaluation frames. Yellow boxes are annotations and green boxes are predictions at validation-fixed operating thresholds. TRACE-R labels each frame as native pass-through or native-plus-restored evidence.